# HW #8

ALGEBRAIC NUMBER THEORY, GU4043; INSTRUCTOR: GYUJIN OH

**Reading Homework.** Try Exercises 4.3, 4.4, 4.5 in the textbook. Read its solutions in the back.

**Question 1.** Let $A$ be an integral domain, and let $|\cdot| : A \to \mathbb{R}_{\geq 0}$ be a non-archimedean absolute value.

(1) Define
$$A^\circ := \{a \in A \ : \ |a| \leq 1\},$$
$$A^{\circ\circ} := \{a \in A \ : \ |a| < 1\}.$$
Show that $A^\circ$ is a subring of $A$, and $A^{\circ\circ} \subset A^\circ$ is a prime ideal.

(2) On $A[X]$, define the **Gauss norm** $|\cdot|_{\text{Gauss}} : A[X] \to \mathbb{R}_{\geq 0}$ as
$$\left|a_n X^n + a_{n-1}X^{n-1} + \cdots + a_0\right|_{\text{Gauss}} := \max(|a_n|, \ |a_{n-1}|, \ \cdots, \ |a_0|).$$
Show that the Gauss norm is a non-archimedean absolute value on $A[X]$.

**Hint.** By taking $\mathrm{Frac}(A)$, we can assume that $A$ is a field. The only nontrivial axiom is $|f(X)g(X)|_{\text{Gauss}} = |f(X)|_{\text{Gauss}}|g(X)|_{\text{Gauss}}$ for $f(X), g(X) \in A[X]$. It is easy to show $|f(X)g(X)|_{\text{Gauss}} \leq |f(X)|_{\text{Gauss}}|g(X)|_{\text{Gauss}}$. If $|f(X)g(X)|_{\text{Gauss}} < |f(X)|_{\text{Gauss}}|g(X)|_{\text{Gauss}}$ for some $f(X), g(X) \in A[X]$, after multiplying by an appropriate element in $A$, one can assume, without loss of generality, that $|f(X)|_{\text{Gauss}} = |g(X)|_{\text{Gauss}} = 1$. This implies that
$$f(X), g(X) \in A^\circ[X], \ f(X)g(X) \in A^{\circ\circ}[X]$$
Deduce a contradiction by showing that $A^\circ[X]/A^{\circ\circ}[X] \cong (A^\circ/A^{\circ\circ})[X]$ and using (1).

(3) Use (2) show the following[1].

> **Gauss's lemma**. Let $A$ be a UFD. A polynomial $f(X) \in A[X]$ is called **primitive** if the gcd of its coefficients is 1. Then, the product of primitive polynomials is primitive.

**Hint.** For an irreducible element $\pi \in A$, there is a normalized discrete valuation $v_\pi$ on $A$. Taking $0 < \epsilon < 1$, one can define a non-archimedean absolute value $|\cdot|_\pi$ on $A$ by $|x|_\pi := \epsilon^{v_\pi(x)}$.

**Question 2.**

(1) Let $L/K$ be a Galois extension of $p$-adic local fields, and let $|\cdot|$ is a non-archimedean absolute value on $L$. Show that any element $\sigma \in \mathrm{Gal}(L/K)$ is an isometry. In other words, show that $|x - y| = |\sigma(x) - \sigma(y)|$ for any $x, y \in L$.

**Hint.** Use Theorem 4.44.

[1] A special version was stated in Theorem 1.11.

(2) **(Krasner's lemma)** Let $K$ be a $p$-adic local field. Let $f(X) \in K[X]$ be an irreducible separable monic polynomial, and let $L/K$ be a separable extension of local fields containing the splitting field of $f(X)$; in other words, $f(X)$ factorizes in $L[X]$ as

$$f(X) = \prod_{i=1}^{n} (X - \alpha_i), \quad \alpha_i \in L.$$

Let $|\cdot|$ be a non-archimedean absolute value on $L$. Let $\beta \in L$ be such that

$$|\beta - \alpha_1| < |\alpha_j - \alpha_1| \quad \text{for all } j \neq 1.$$

Show that $K(\alpha_1) \subset K(\beta)$.

**Hint.** Enlarge $L$ so that $L/K$ is Galois. If $K(\alpha_1) \not\subset K(\beta)$, there should be $\sigma \in \mathrm{Gal}(L/K)$ such that $\sigma(\alpha_1) \neq \alpha_1$ and $\sigma(\beta) = \beta$. Use (1) to deduce a contradiction.

(3) Let $K$ be a $p$-adic local field, and $|\cdot|$ be a non-archimedean absolute value on $K$ (or its unique extension on any finite extension of $K$). Let $L/K$ be a finite Galois extension. Show that there exists $0 < c_{L/K} \leq 1$ (depending only on $L$, $K$, $|\cdot|$) such that the following holds.

**Lemma** (Fontaine). Let $E/K$ be any Galois extension, and let $0 < \epsilon < c_{L/K}$. Define

$$\mathfrak{m}_E^{\epsilon} = \{x \in \mathcal{O}_E \ : \ |x| \leq \epsilon\}.$$

If there exists an $\mathcal{O}_K$-algebra homomorphism $\mathcal{O}_L \to \mathcal{O}_E/\mathfrak{m}_E^{\epsilon}$, there exists an $\mathcal{O}_K$-algebra homomorphism $\mathcal{O}_L \to \mathcal{O}_E$.

**Hint.** Show first that $\mathcal{O}_L = \mathcal{O}_K[\alpha]$ for some $\alpha \in \mathcal{O}_L$ with $|\alpha| < 1$. Let $f(X) \in \mathcal{O}_K[X]$ be the minimal polynomial of $\alpha$ over $\mathcal{O}_K$. What the problem is asking is, if there is $\beta \in \mathcal{O}_E$ such that $|f(\beta)| \leq \epsilon$, whether there is $\beta' \in \mathcal{O}_E$ such that $f(\beta') = 0$. Use (2).